

CSharPN

A Code-First Framework for Coloured Petri Net Modelling
with C# as Inscription Language

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The gap

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What if a CPN model were just a C# file in your repo?

CPNs in 30 seconds

Two extensions used in this talk:

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- **Timed CPNs**: tokens carry timestamps, global clock advances
- **Hierarchical CPNs**: substitution transitions + port places (In, Out, I/O)

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Open source (MIT) — github.com/simontjell/CSharPN.public

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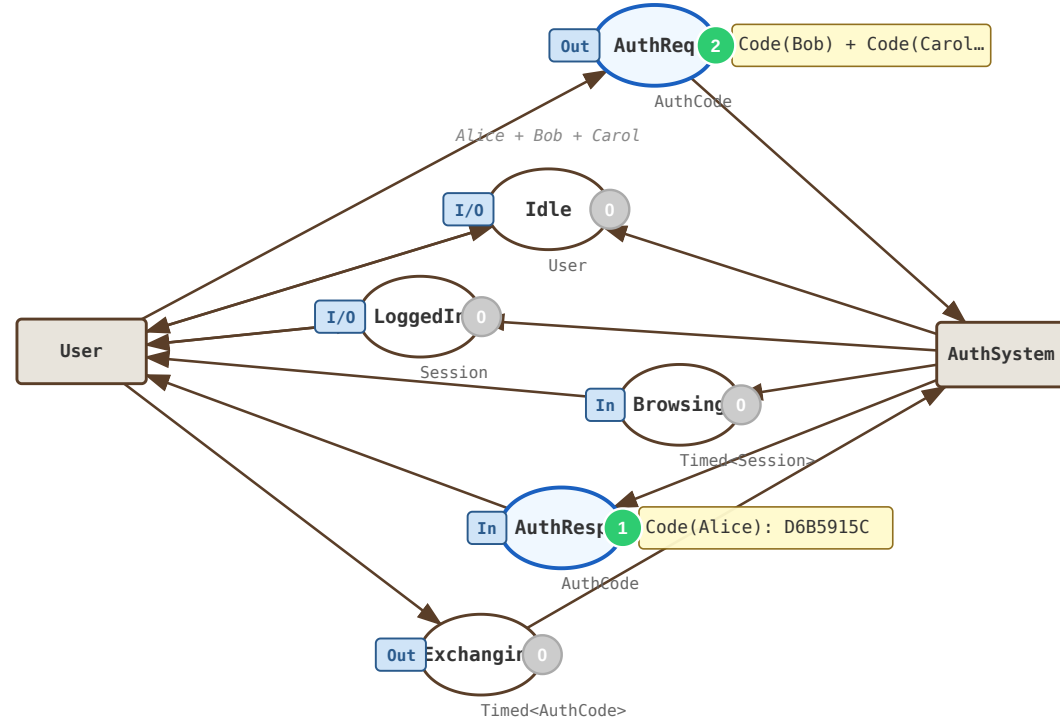
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- Interactive **Blazor visualiser** embedded in VS Code

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Running example: OAuth2 authoriz

Exercises all three features:

Top page: shared places link the two substitution transitions.

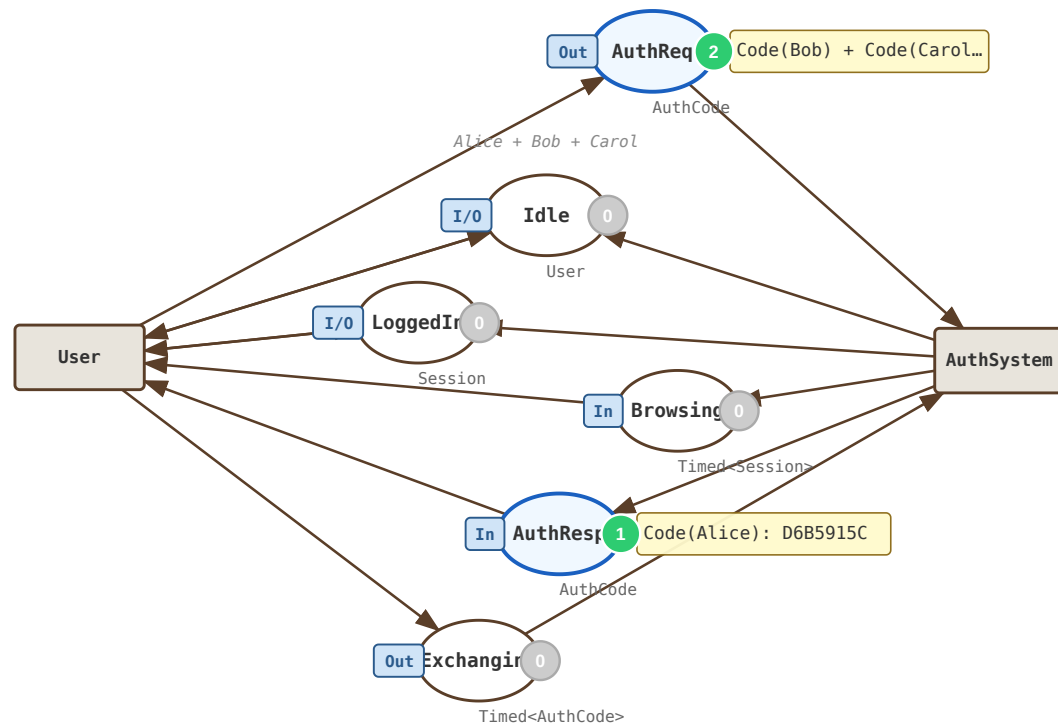


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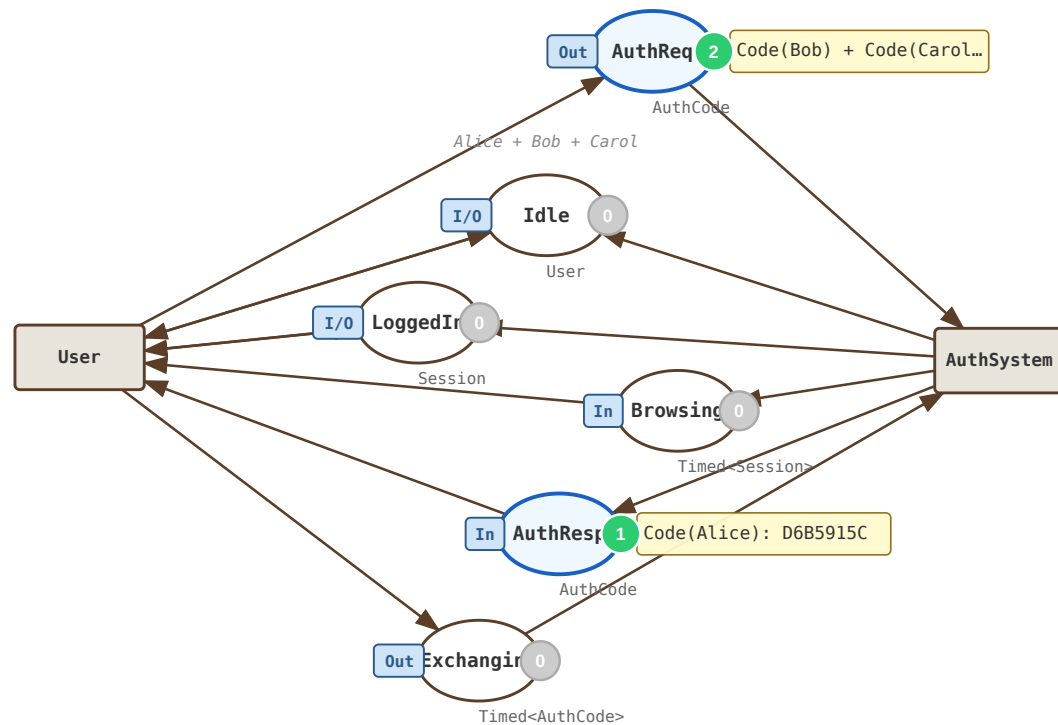


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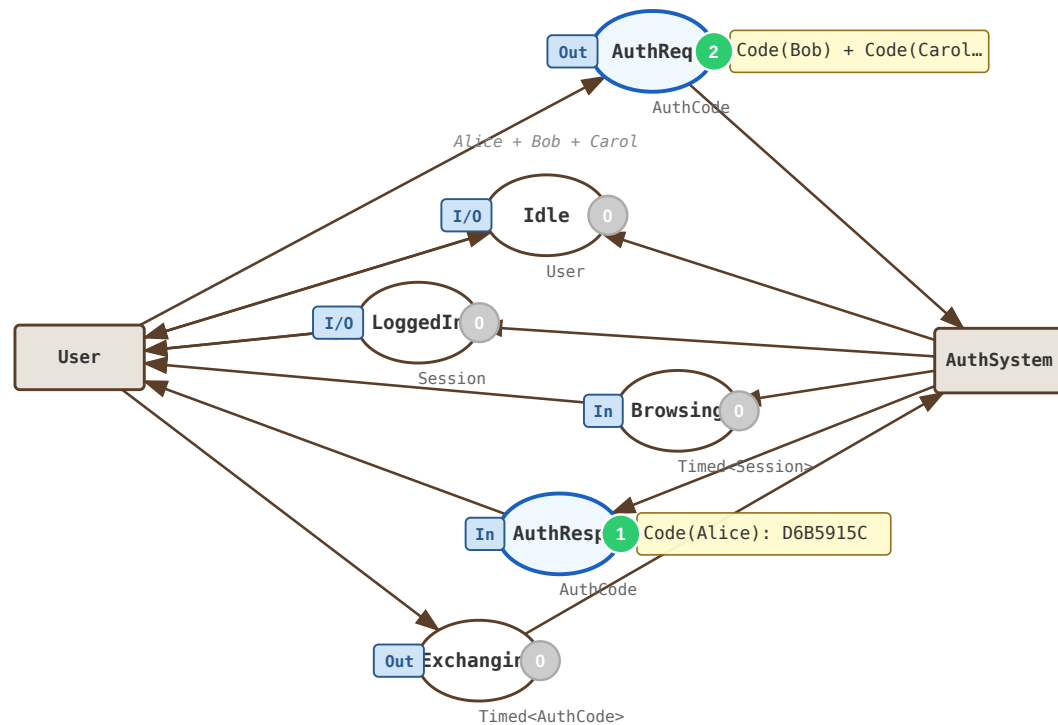


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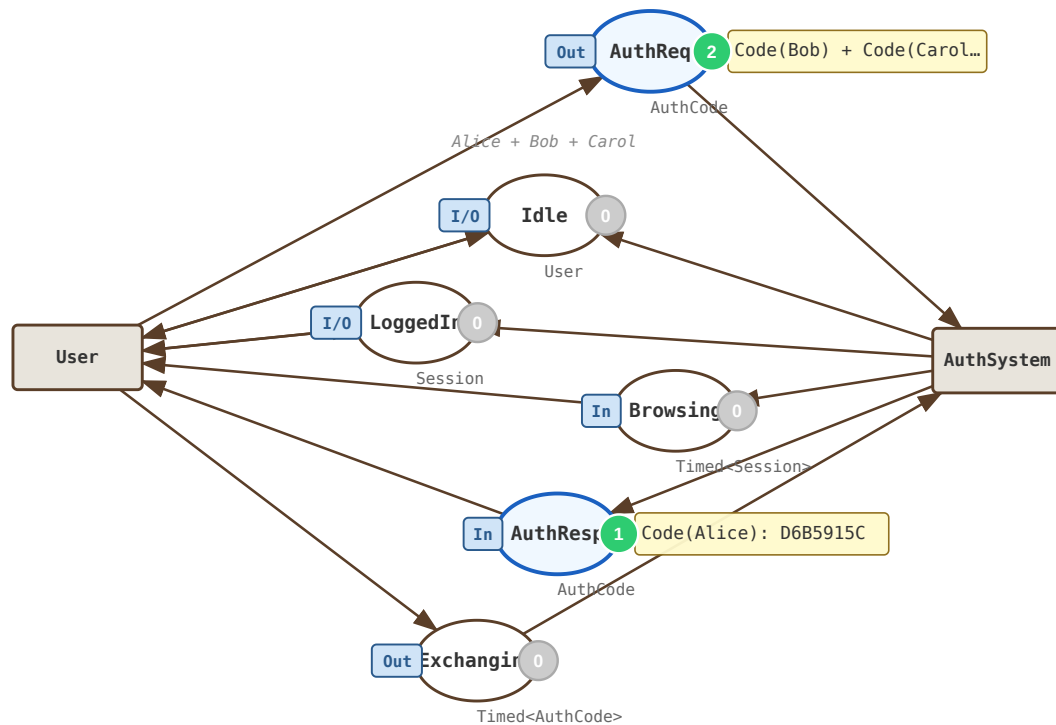


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- Two sub-pages: User & AuthSystem

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Colour sets are just C# types

Any `notnull, IEquatable<T>` type works — primitives, enums, records, classes.

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public record User(int Id, string Name) {  
    public string CsrfsState => $"s{Id}";  
    public int    BrowseTime => Id * 5;  
}  
  
public record AuthCode(User User, string Code, string State) {  
    public bool ValidCsrfs(User u) => State == u.CsrfsState;  
    public static AuthCode Create(User u, int seq) {  
        var hash = SHA256.HashData(Encoding.UTF8.GetBytes($"{u.Id}:{seq}"));  
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Places & multisets

```
// inside the model constructor
var authReq = AddPlace<AuthCode>("AuthReq");           // empty initial marking

var idle    = AddPlace("Idle", Multiset.Of(
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var exchg   = AddTimedPlace<AuthCode>("Exchanging");    // timed place
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var u = new Var<User>("u");

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    .Input(idle, u).Output(waiting, u)
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    .Build();

AddTransition("ReceiveCode")
    .Input(authResp, c).Input(waiting, u)
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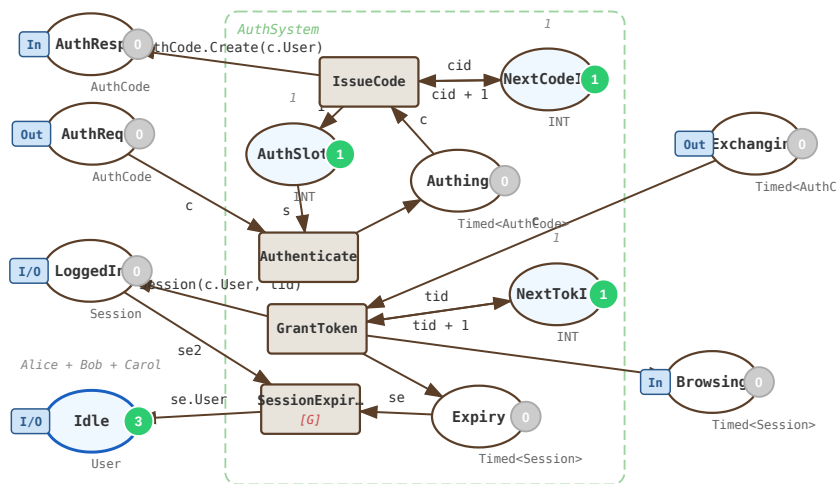
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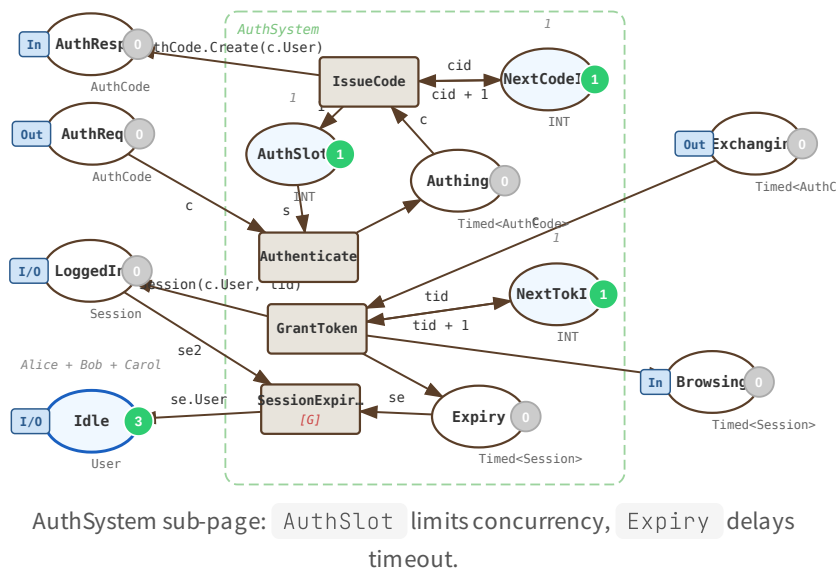
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class UserPage : CpnPage {
    public UserPage(Place<User> idle,
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// assembly
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AuthSystem sub-page: **AuthSlot** limits concurrency, **Expiry** delays timeout.

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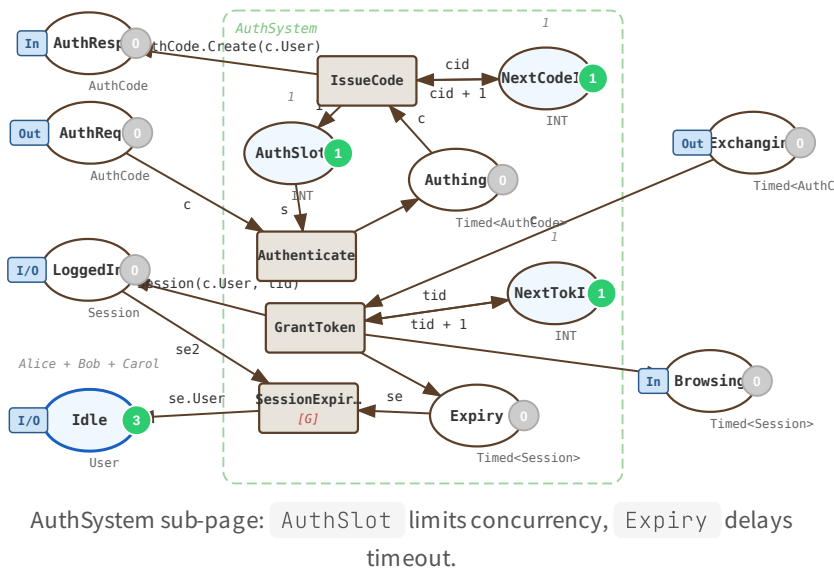
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- A sub-page is a `CpnPage` subclass

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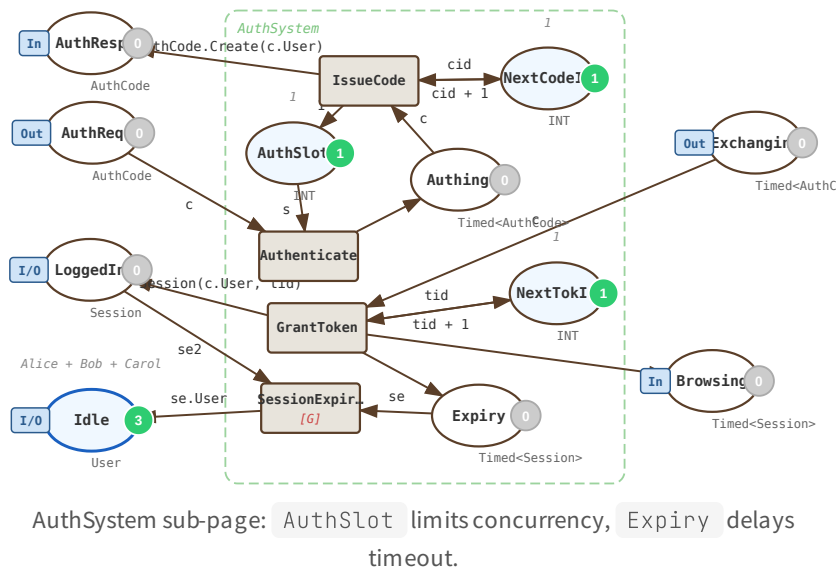
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```



- A sub-page is a `CpnPage` **subclass**
- Ports use `In()` / `Out()` / `InOut()` — CPN Tools style

Hierarchical CPNs

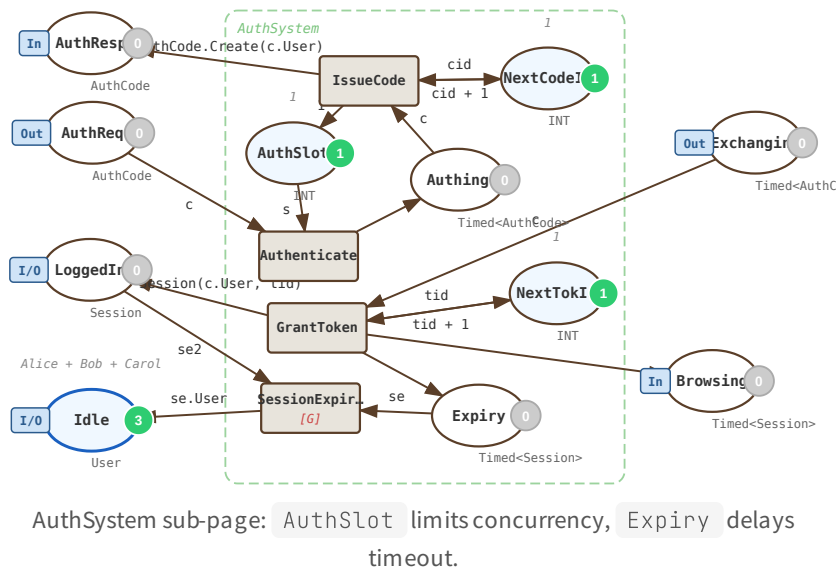
```
class UserPage : CpnPage {  
    public UserPage(Place<User> idle,  
        Place<AuthCode> authReq, ...)  
        : base("User") {  
        InOut(idle);  
        Out(authReq); In(authResp);  
        Out(exchg);  
        InOut(loggedIn); In(browsing);  
        var waiting = AddPlace<User>("Waiting");  
        // Login, ReceiveCode, Logout  
    }  
}  
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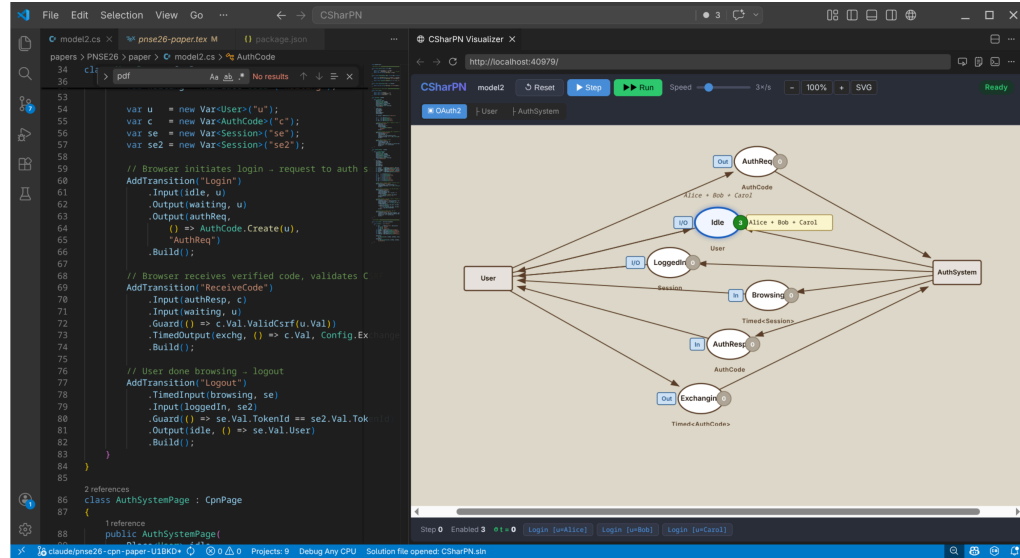
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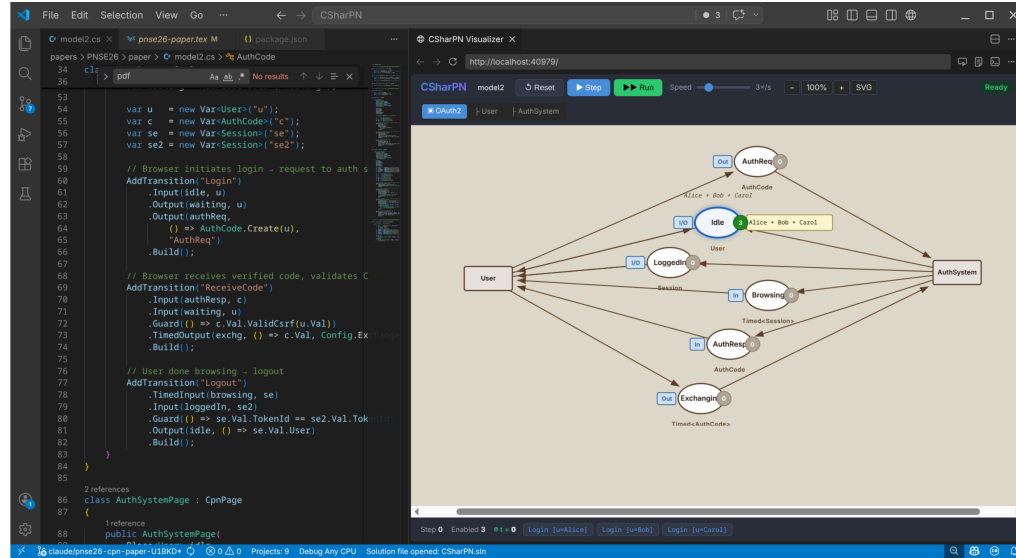
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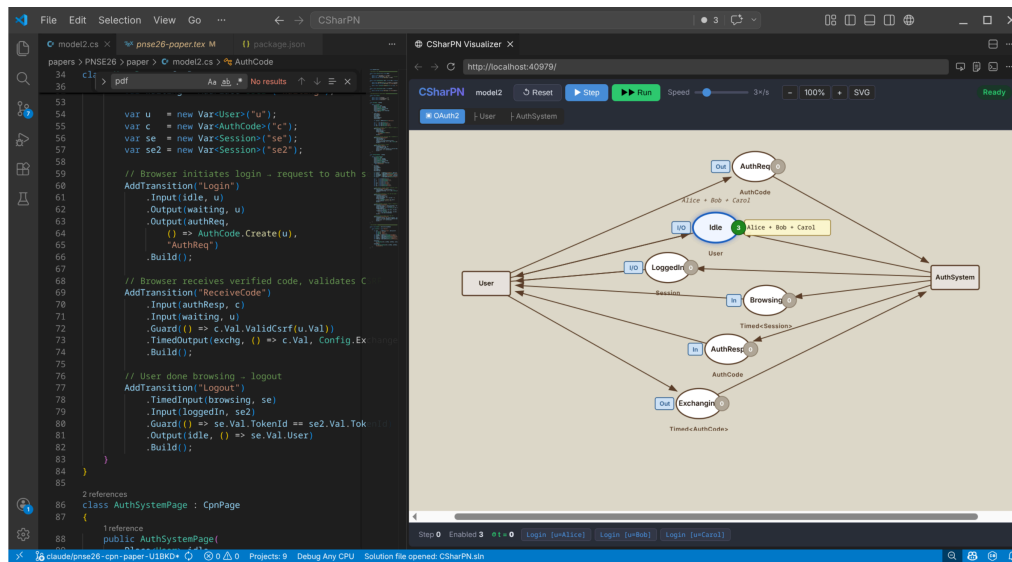
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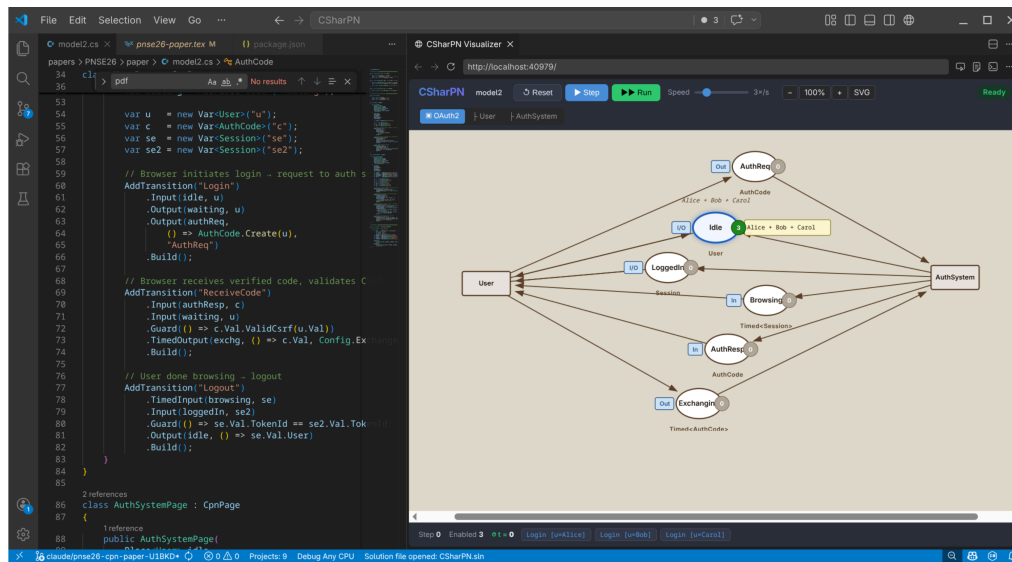
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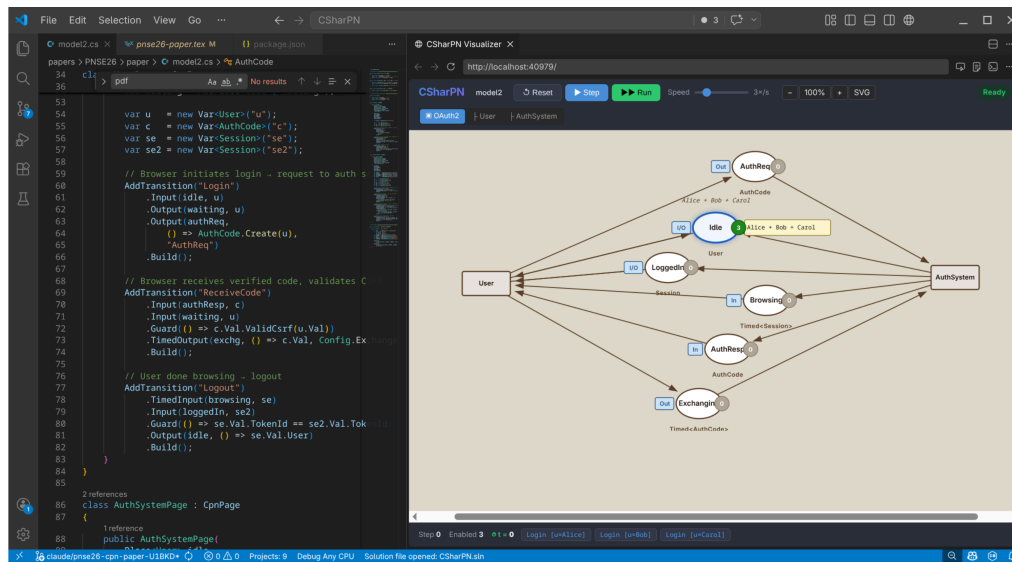
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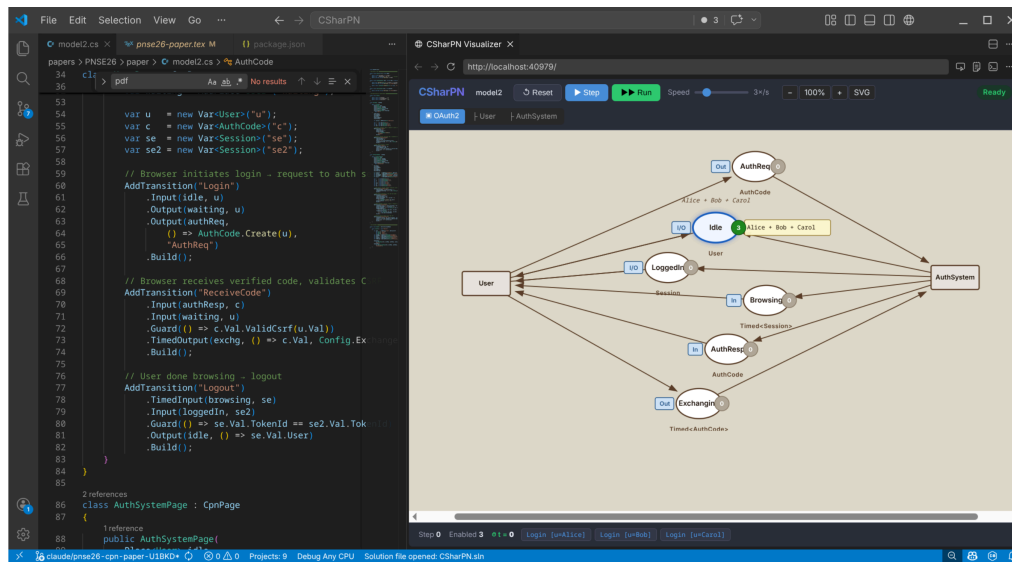
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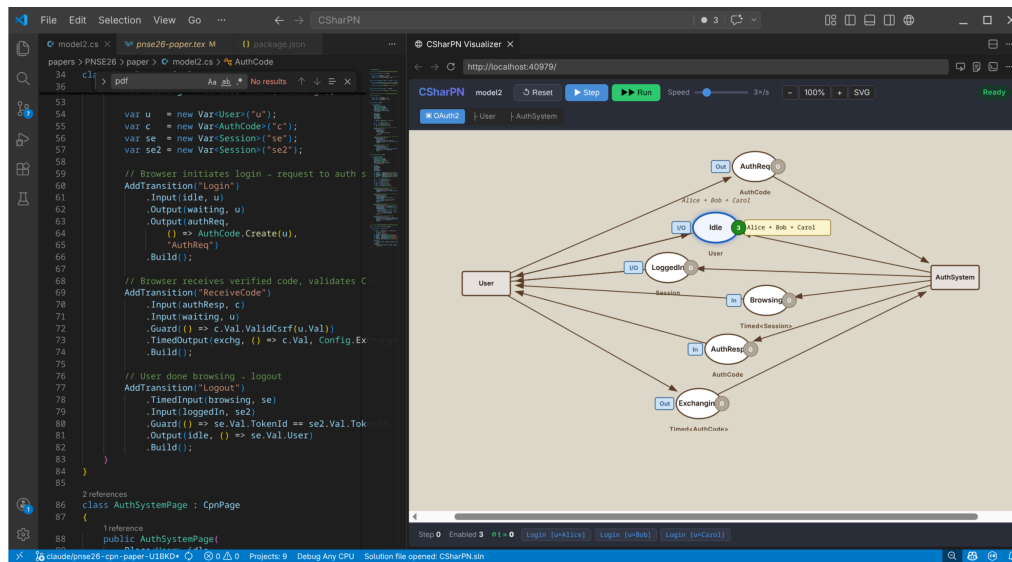
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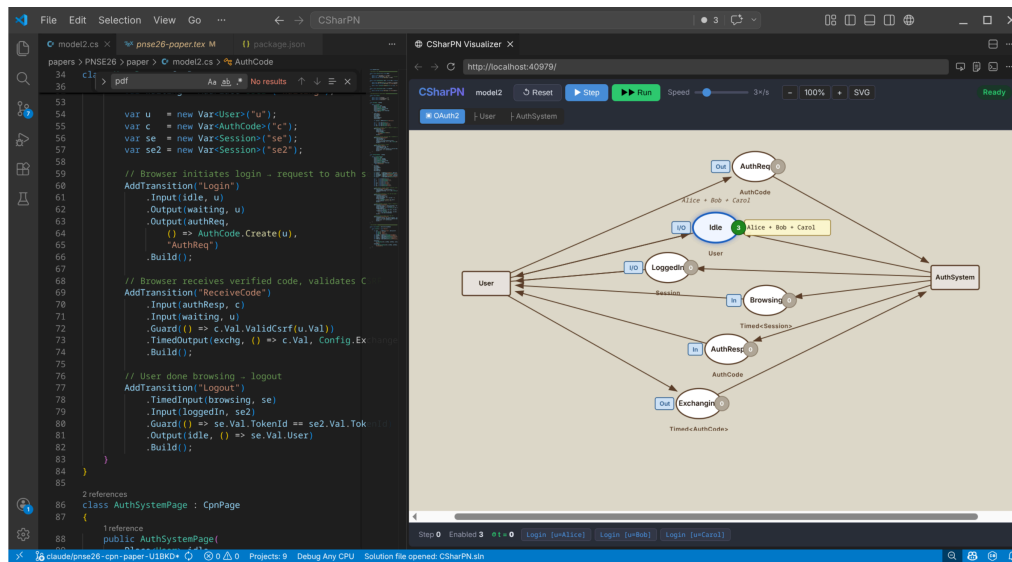
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- **Hot reload**: save .cs → recompile → reload, marking preserved



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key insight

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No code-generation step. No generated artefact, no model/implementation sync problem — the gap is zero by construction. (Caveat:

only when the deployment target is .NET.)

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Takeaway: when the inscription language is C#, the full power of a mature programming language — type system, libraries, tooling — is available to the CPN modeller at no extra cost.

Thank you

Questions?



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github.com/simontjell/CSharpN.public

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